

CASTERTON, Australia

WMO: 958250

Lat: 37.5831S

Lon: 141.3339E

Elev: 430

StdP: 14.47

Time Zone: 10.00 (AUS)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.9	35.9	31.3	26.0	50.7	33.3	28.3	46.3	23.7	53.8	21.6	52.1	2.2	60	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.7	95.8	65.2	90.8	64.6	86.3	63.8	68.9	83.5	67.2	82.1	65.6	80.7	12.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	94.4	71.7	62.8	86.8	70.1	60.8	81.0	68.7	33.3	82.4	31.9	81.4	30.7	80.8	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.9	17.3	DB	30.1	104.9	2.0	4.3	28.7	108.0	27.5	110.5	26.4	112.9	25.0	116.0	(4)
(5)			WB	29.6	72.7	2.0	2.2	28.2	74.3	27.0	75.6	25.9	76.8	24.5	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	67.4	67.0	63.9	58.0	52.8	48.6	48.1	49.1	51.4	55.1	59.3	62.7		
	DBStd	9.10	7.86	7.22	7.45	5.88	4.73	3.61	3.63	4.12	4.71	6.39	7.25	7.36		
	HDD50	295	0	0	0	3	22	68	82	66	34	17	3	0		
	HDD65	3411	59	57	114	225	378	492	525	492	408	318	206	137		
	CDD50	2812	538	475	432	243	110	26	22	40	76	174	282	394		
	CDD65	451	132	112	81	14	1	0	0	0	0	10	36	65		
	CDH74	5893	1820	1265	946	160	2	0	0	0	3	166	514	1017		
	CDH80	2679	951	579	412	39	0	0	0	0	0	42	184	472		
(14) Wind	WSAvg	8.7	9.2	9.3	8.7	7.5	7.7	6.9	8.9	9.2	9.2	9.1	9.1	9.4		
(15) Precipitation	PrecAvg	25.9	1.1	0.9	1.5	1.7	2.4	2.8	3.4	3.4	3.0	2.4	1.9	1.5		
	PrecMax	35.4	3.5	3.7	4.7	5.1	6.2	6.2	7.4	6.5	5.6	5.8	5.0	5.4		
	PrecMin	14.6	0.2	0.0	0.2	0.2	0.3	0.5	1.3	0.5	1.1	0.5	0.4	0.2		
	PrecStd	4.7	0.7	0.8	1.1	1.1	1.3	1.2	1.3	1.3	1.1	1.3	0.9	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.2	97.1	97.8	84.8	70.8	61.1	60.5	67.8	72.7	85.4	90.7	99.1		
		MCWB	67.4	65.8	64.2	61.5	57.8	54.6	55.0	55.3	59.5	61.4	65.1	65.3		
	2%	DB	97.5	92.6	89.6	78.3	66.5	58.4	57.4	63.3	68.0	78.6	85.6	91.8		
		MCWB	65.6	65.5	64.0	59.6	56.2	53.5	52.9	53.8	56.8	60.3	64.2	63.0		
	5%	DB	90.4	87.6	84.1	73.6	63.3	56.7	55.6	59.1	64.1	73.0	80.2	85.0		
		MCWB	64.6	65.0	62.8	58.0	56.1	52.8	51.7	52.6	55.0	58.4	63.2	63.1		
	10%	DB	84.1	81.9	78.6	69.0	60.7	55.0	53.9	56.2	60.9	68.1	74.3	78.9		
		MCWB	63.5	63.6	61.6	57.1	55.1	51.8	50.5	51.0	53.7	57.1	61.3	61.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.0	71.4	68.6	63.9	60.3	56.9	56.8	57.8	61.0	66.1	70.3	70.0		
		MCDB	88.8	87.1	79.2	75.2	65.1	58.6	58.2	63.2	67.4	78.5	84.1	84.1		
	2%	WB	68.7	68.0	66.2	61.8	58.6	54.9	54.4	55.0	58.3	61.9	67.2	66.6		
		MCDB	84.4	84.7	82.1	71.5	63.2	56.8	56.0	60.5	65.0	73.1	79.7	81.9		
	5%	WB	66.9	66.2	64.5	60.2	57.3	53.5	52.7	53.1	56.1	60.0	64.8	64.5		
		MCDB	83.0	80.9	78.7	68.8	61.3	55.6	54.5	57.8	62.5	70.9	76.7	79.7		
	10%	WB	64.9	64.5	63.0	58.7	55.9	52.3	51.3	51.8	54.2	57.8	62.0	62.7		
		MCDB	81.5	79.2	75.8	66.2	59.7	54.5	53.3	55.5	60.0	66.8	73.1	77.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	26.7	23.8	22.1	18.7	13.8	13.1	11.9	13.8	16.5	20.4	22.3	25.3		
		MCDBR	37.1	32.9	31.9	27.4	16.8	15.1	12.8	17.8	21.1	29.8	32.4	37.0		
		MCWBR	11.3	10.5	10.5	11.6	9.8	11.2	9.5	11.0	12.4	14.4	14.3	12.5		
	5% WB	MCDBR	28.7	26.9	25.5	20.8	14.2	13.2	11.9	15.4	19.0	25.4	27.5	28.6		
		MCWBR	11.6	10.6	10.1	10.4	9.2	10.4	9.5	10.3	12.2	13.9	14.2	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.338	0.333	0.321	0.310	0.304	0.298	0.298	0.312	0.333	0.336	0.335	0.335		
	taud		2.503	2.533	2.564	2.578	2.571	2.576	2.572	2.519	2.449	2.447	2.483	2.503		
	Ebn at Noon		315	310	301	285	268	260	267	280	292	305	315	318		
	Edh at Noon		36	33	30	26	22	21	22	27	33	36	36	36		
(44) All-Sky Solar Radiation	RadAvg		2318	2010	1567	1088	723	588	639	909	1266	1683	2007	2249		
	RadStd		139	99	72	72	37	45	37	48	98	143	139	142		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (24 neighbors)		+0.70	N/A	N/A	+1.62	N/A	N/A	N/A	-173	+238	+70	(47)

Nomenclature: See separate page